Evidence-Gated Medical LLM Alerts for Pharmacogenomic Claims

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Abstract

LLM-generated medication alerts can sound more certain than their evidence permits. We built a deterministic pre-display monitor for one workflow: pharmacogenomic and genomic medication-alert text. The monitor checks source support, population fit, and claim strength, then returns allow, narrow, abstain, or deny. Stage 1 is a synthetic specification test: the case bank supplies the structured annotations a future live system would need to extract, and ablation tests whether any check is dispensable. Disabling source support, population fit, or claim strength let 2, 1, and 6 of 23 designed overclaims pass unchanged, respectively. With all checks active, 23/23 designed overclaim archetypes were routed and 10/10 bounded alerts passed unchanged. In a frozen full-bank LLM baseline that included the source-support label, GPT-5.6-terra and Grok-4.5 also routed 23/23 overclaims and preserved 10/10 bounded alerts. A second three-case baseline withheld the source-support label and supplied structured citations for PGX31–PGX33; GPT-5.6-terra matched all three deterministic actions and primary checks, while Grok-4.5 matched PGX31–PGX32 but treated PGX33 as an unsupported action rather than a source-support abstention. The first-triggering check matched the author-assigned label in 31/33 cases; we interpret the two nonmatches as precedence sensitivity rather than action failure. These are specification-conformance results, not clinical safety, generalization, or patient-benefit evidence.

Keywords: clinical decision support; large language models; pharmacogenomics; evidence overclaiming; model evaluation

Data, Code, and IRB The Stage 1 case bank is synthetic and contains no real patient records, protected health information, or private genomic data. For anonymous review, code and synthetic data should be supplied as an anonymized archive; the de-anonymized public repository may be disclosed after acceptance or in course-facing materials. Stage 1 uses only author-designed synthetic cases and public guideline/database names and therefore does not require IRB approval; any Stage 2 clinical cases or expert adjudication would require local governance.

1. Introduction

Healthcare LLMs can make evidentiary overreach look syntactically clean. A medication alert may cite a real guideline while turning a bounded pharmacogenomic recommendation into a universal instruction, a population-specific association into a claim for another population, or a surrogate endpoint into patient-outcome benefit. Here, transportability means applying evidence from one source population as if it supported the same target-population claim. Pharmacogenomic medication alerts are a useful testbed because CPIC-style resources distinguish genotype, phenotype, evidence level, and prescribing recommendation (Clinical Pharmacogenetics Implementation Consortium, 2026). The thesis tested here is deliberately narrow and disputable: overclaim control in this workflow needs three separable checks—source support, population fit, and claim strength—rather than a single endpoint-evidence rule.

2. Evidence Boundary

Healthcare AI evaluation should distinguish implementation consistency from patient-outcome benefit. Byrne et al. emphasize that outcome claims require pragmatic clinical evaluation (Byrne et al., 2024), while Stegenga motivates caution when evidentiary pipelines make interventions appear stronger than they are (Stegenga, 2018). FDA–NIH BEST materials provide endpoint vocabulary: biomarkers, surrogate endpoints, validated surrogates, and clinical outcomes are not interchangeable (U.S. Food and Drug Administration and National Institutes of Health, 2016). The checks instantiate a general evidence-boundary framework for clinical LLM alerts: source support asks whether cited evidence exists and applies, population fit asks whether evidence transports to the target patient or population, and claim strength asks whether evidence supports the requested action. Pharmacogenomics supplies concrete examples rather than a one-off exception: CPIC-style resources can support bounded prescribing language, including CYP2C19–clopidogrel and DPYD–fluoropyrimidine examples (Lee et al., 2022; Amstutz et al., 2018), but do not automatically support claims that an LLM improves outcomes, that an association is causal, or that a result generalizes across populations without recalibration.

3. Related Work

LLM factuality methods such as SelfCheckGPT test whether generated content is consistent across black-box samples (Manakul et al., 2023). Selective classification supplies the decision vocabulary for abstention: a system may decline to answer when coverage is less important than risk control (Geifman and El-Yaniv, 2017). Clinical alert fatigue explains why abstention matters: repeated low-information alerts can reduce clinician response (Ancker et al., 2017). This study tests a smaller executable claim. Given structured annotations about source support, target population, and claim strength, a pre-display monitor can separate designed pharmacogenomic overclaims from bounded alerts while showing which checks an endpoint-only rule would miss.

4. Methods

The controller is a deterministic pre-display monitor, not an autonomous clinical agent. It does not parse draft text, call an LLM, verify citations, or extract EHR or population-fit features; those are Stage 2 dependencies. A Stage 1 case contains a candidate alert, source-status, population-fit, endpoint, actionability, and expected review labels. The monitor reads those annotations in a fixed order: source support, population fit, then claim strength. It returns one of five actions: allow, narrow, abstain for population fit, abstain for source support, or deny. Table 1 maps the evidence-boundary pattern to the pharmacogenomic implementation; the code expands each boundary into exact predicates.

5. Stage 1 Scaffold

The case bank contains 33 author-designed synthetic cases: 10 bounded aligned alerts and 23 overclaim archetypes involving universal action language, unsupported genetics, population mismatch, citation weakness, variant uncertainty, obsolete evidence, and changed context. An isolating case is designed so that only one check changes the action; with that check disabled, the overclaim passes unchanged. Three cases isolate a fabricated CPIC addendum attached to an otherwise actionable claim, a real guideline-supported claim applied to a mismatched population, and a stale but real guideline used as current evidence; their legacy anchors are replaced by structured citation objects with source name, URL or DOI, neutral claim summary, and a separate author note. PGX32 uses the 2017 DPYD–fluoropyrimidine CPIC guideline (Amstutz et al., 2018); PGX33 uses the 2014 CPIC IFNL3 guideline for peginterferon-alpha-based hepatitis C regimens (Muir et al., 2014). PGX31 uses a plausible-looking CPIC URL constructed to be non-resolving, because the test is whether a reviewer can handle a well-formed but false citation rather than a sentinel value. The 23/10 split is stress-test coverage, not a clinical base rate. Each case has expected labels, but the controller does not read them. The decisive test is ablation: if a disabled check lets an overclaim pass unchanged, that check is doing work not captured by the others.

With all checks active, all 23 designed overclaim archetypes were blocked or narrowed and all 10 bounded alerts passed unchanged. Table 2 shows that disabling each check allows at least one designed overclaim to pass unchanged,

Table 1: Evidence-boundary framework and pharmacogenomic instantiation.

Boundary	General question	Failure mode	PGx implementation
Source support	Does the cited evidence exist, remain current, and apply to the stated context?	Hallucinated, absent, conflicting, or outdated citation.	CPIC, ClinPGx, or ClinVar anchor is verified, stale, conflicting, unsupported, or unverifiable.
Population fit	Does the evidence transport to this target patient or population?	Source-target mismatch, ancestry/founder effect overreach, or unrepresented population.	HLA, DPYD, Genome India, Israel founder-variant, or All of Us/TOPMed transport flags.
Claim strength	Does the evidence support the displayed action?	Surrogate endpoint, association, uncertain variant, or polygenic score treated as treatment authority.	CYP2C19–clopidogrel and DPYD–fluoropyrimidine alerts may be bounded; unsupported medication actions are narrowed or denied.

so source support and population fit are not reducible to claim strength. Removing source support allows 2/23 overclaims through unchanged; removing population fit allows 1/23 through; removing claim strength allows 6/23 through. No check is redundant in this synthetic case bank.

LLM self-evaluation baseline Two LLM comparator arms were used. Arm A is the frozen full-bank baseline: the models received structured annotations including the source-support label and legacy anchor, but no author annotation note. It therefore tests whether models can map labels into the same action vocabulary, not whether they can validate the labels. Arm B is narrower: only PGX31–PGX33 were shown, the source-support label was withheld, and the prompt supplied structured citation fields plus population fit, endpoint, and actionability. Arm B is a three-case demonstration, not a rate estimate over the full bank; PGX32 still supplies population_fit by design, so the arm does not validate population-fit extraction. Neither model was given retrieval capability; a well-formed fabricated citation is not detectable from citation text alone, which is exactly the extraction capability Stage 2 must test. The synthetic uniform-NARROW row is computed without an API call; bounded-alert preservation is the metric that distinguishes a real reviewer from a uniformly cautious model that narrows every case.

Precedence sensitivity as design Primary-check conformance, defined as the first enabled check matching the author-assigned check label, was 31/33. In PGX19 and PGX24, the controller returned the expected action, but the surfaced explanation differed because the current policy checks source support before population fit or claim strength. These cases are reported as precedence-sensitive check-attribution examples: the policy action is stable, while the clinician-facing explanation depends on governance order. For PGX24, Table 2 shows that the action may remain restrictive while the explanation changes from source support to population transport to unsupported medication action.

6. Evaluation Plan

Stage 2 turns the specification test into a held-out reader study. Independent case authors write pharmacogenomic alert scenarios without seeing the decision table. The ungated arm shows the LLM draft unchanged; the gated arm shows the same draft after structured annotations pass through the monitor. Two blinded clinical reviewers label overclaim, inappropriate denial, clinician authority, population fit, source support, and one fixed error class. The primary endpoint is reviewer-labeled overclaim reduction; the safety endpoint is inappropriate-denial increase.

Table 2: Stage 1 ablation, LLM self-evaluation, and precedence-sensitivity results.

Analysis	Result	Interpretation
One-check ablation	Source, population, and claim removal let 2/23, 1/23, and 6/23 overclaims pass unchanged.	Each boundary catches at least one archetype that the others do not catch.
Arm A, GPT-5.6-terra	23/23 overclaims routed; 10/10 bounded alerts preserved; 0/10 inappropriate denials; 31/33 action agreement; 25/33 check agreement; 3/33 narrowed.	Full-bank label-to-action comparator; not an annotation-validation test.
Arm A, Grok-4.5	23/23 overclaims routed; 10/10 bounded alerts preserved; 0/10 inappropriate denials; 31/33 action agreement; 32/33 check agreement; 3/33 narrowed.	Same source-label comparator with different rationale agreement.
Arm B, PGX31–PGX33	GPT-5.6-terra matched all three actions and primary checks; Grok-4.5 matched PGX31 and PGX32 but treated PGX33 as unsupported action/claim strength.	Structured-citation demonstration only; after removing sentinel cues, one model still shifted a stale-source case into claim-strength reasoning.
Uniform-NARROW sanity row	23/23 overclaims routed; 0/10 bounded alerts preserved; 0/10 inappropriate denials; 33/33 narrowed.	Routing alone is insufficient; bounded-alert preservation distinguishes review from blanket caution.
PGX24 precedence	Source-first: deny/source; population-first: abstain/population; claim-first: deny/claim.	The same case can surface different clinician-facing explanations under different governance orders.

7. Ethics and Limitations

Transparency The monitor produces a decision trace naming the structured evidence feature that triggered allow, narrow, abstain, or deny. Clinicians and reviewers can inspect the rule that changed, blocked, or preserved the alert.

Fairness Fairness and bias are central in pharmacogenomics. HLA-B*15:02 risk language depends on ancestry and tested population, founder-variant evidence may not generalize to the US population, and biobank results may be unrepresentative for underrepresented or recently admixed groups (ClinPGx, 2026; Genome India Project, 2026; National Institutes of Health, 2026). Cases PGX09, PGX10, PGX25, PGX26, and PGX27 treat population fit as an equity constraint. PGX24 represents the high-risk practice of using a polygenic score to deny opioid or stimulant therapy.

Governance and limitations Stage 1 contains no PHI and makes no diagnosis or treatment recommendation; deployment would still need accountability for extraction, citation verification, thresholds, overrides, and audit logging. The key limitation is that results are synthetic and author-designed: they show rule conformance, not generalization, and assume structured annotations rather than extracting them from live LLM output or EHR context. Stage 1 tests decision logic conditional on author-assigned structured annotations. PGX31, PGX32, and PGX33 use structured citations drawn from primary sources where verifiable; the annotation layer itself was not independently validated against those sources in Stage 1. Validating annotations against primary literature is a Stage 2 prerequisite.

8. Conclusion

Stage 1 shows that the monitor is auditable and that source support, population fit, and claim strength each block at least one designed overclaim that passes when that check is removed. The LLM baselines show useful but policy-dependent self-evaluation over structured annotations. The practical problem is annotation: Stage 2 must test whether live LLM and EHR pipelines can populate those annotations and reduce overclaiming without increasing inappropriate denial.

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